

Course Title	Engineering Mathematics II		
Course Code	ED 121	Credit Structure	3-0-1-4 (L-T-P-C)
Category	Core	Semester	Winter Semester (AY 25-26)
Program	B.Tech		
Prerequisites	NA		
Course Objectives/ Brief Course Description	- Understand and define basics of probability including sample space, events, random variable, probability distribution function, probability mass function etc. - Learning about different probability distributions including Binomial, Poisson, Geometric, Uniform, Normal, Exponential, Chi-Square distributions. - Knowledge about expectation and variance. - Central limit theorem. Learning about applications of probability - Learning about basic statistical measures including mean standard deviation. - Knowledge about hypothesis testing and significance level. - Regression, least-square estimators and applications. - Knowledge of basis information theory - Learning about entropy, joint entropy, conditional entropy etc. - Applications of information theory.		
Evaluation/ Grading Policy	Homework -15% Midterm Test 1-20% Midterm Test 2-20% Final Test -30 % Project- 15%		
Course Materials/ References	Lecture notes. Lecture notes/ class notes will be posted and made available for students after the lecture. Books Introductory Statistics, Sheldon Ross, Elsevier Probability and Statistics for Engineers and Scientists by Sheldon Ross, Fourth Edition Elements of Information Theory by Thomas Cover and Joy Thomas, Second Edition.		

Detailed Course Content	Probability: Sample Space, Events, Axioms of Probability, Conditional Probabilities, Independence of Events, Bayes' Theorem, Random Variables, Types of Random Variables, Jointly Distributed Random Variables, Expectation, Variance, Sums of Random Variables, Covariance, Markov's and Chebyshev's Inequality, Law of Large numbers, Special Random Variables: Binomial, Poisson, Geometric, Uniform, Normal, Exponential, Chi-Square, Distribution, Sample Mean, Central Limit Theorem, Sample Variance. Statistics: Parameter Estimation: maximum likelihood estimators, interval estimates, Hypothesis Testing: Significance Level, Mean and Variance of Normal Population, Hypothesis Testing: Mean of Poisson Distribution, Hypothesis testing in Bernoulli populations, Regression, least square estimators, distribution of estimators, Goodness of fit tests. Information Theory: Notion of information, entropy, joint entropy, conditional entropy, relative entropy, mutual information, chain rules, Data Compression, Kraft inequality, Optimal codes, Huffman codes, Channel capacity, Channel Coding Theorem, Hamming Codes.

Course Outcome: CO2, CO4

POs-COs Matrix:

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
x	x		x	x							x

APPENDIX: Detailed Course Content (Session-wise/ Module-wise)

Tentative Lecture Schedule

Sl. No.	Description	No. of Lectures
1	Probability Sample Space, Events, Axioms of Probability, Conditional Probabilities, Independence of Events, Bayes' Theorem, Random Variables, Types of Random Variables, Jointly Distributed Random Variables, Expectation, Variance, Sums of Random Variables, Covariance, Markov's and Chebyshev's Inequality, Law of Large numbers, Special Random Variables: Binomial, Poisson, Geometric, Uniform, Normal, Exponential, Chi-Square, Distribution, Sample Mean, Central Limit Theorem, Sample Variance.	20
2	Statistics Parameter Estimation: maximum likelihood estimators, interval estimates, Hypothesis Testing: Significance Level, Mean and Variance of Normal Population, Hypothesis Testing: Mean of Poisson Distribution, Hypothesis testing in Bernoulli populations, Regression, least square estimators, distribution of estimators, Goodness of fit tests.	10
3	Information Theory: Notion of information, entropy, joint entropy, conditional entropy, relative entropy, mutual information, chain rules, Data Compression, Kraft inequality, Optimal codes, Huffman codes, Channel capacity, Channel Coding Theorem, Hamming Codes.	10